# INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD
## Department of Mathematics

**Final Exam** Time : 90 minutes

Date : October 19, 2024 **MA1220 : Calculus II** Max Marks : 40

**Instructions**:

- No specific charts, graphs, log book etc. required.
- Step marking depends on the procedure and the final answer. Hence, it is essential to show the detailed work.
- Answer **all** questions.

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1. Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the $x$-axis and the line $y = x - 2$. [3]

**Solution:** To find the area of the region, we first need to determine the points of intersection between the curves $y = \sqrt{x}$ and $y = x - 2$.

**Solve for Points of Intersection**

We set $y = \sqrt{x}$ equal to $y = x - 2$ to find the points of intersection

$$\begin{aligned} \sqrt{x} &= x - 2 \\ \implies x &= (x-2)^2 \\ \implies x &= x^2 - 4x + 4 \\ \implies x^2 - 5x + 4 &= 0 \end{aligned}$$

Factoring the quadratic equation

$$(x-4)(x-1) = 0$$

Thus, the solutions are $x = 4$ and $x = 1$.

At $x = 1$, $y = x - 2$ gives the value $-1$, which is not in the first quadrant.

Observe that the key points for splitting the region are

- $x = 0$ to $x = 2$, where the region is bounded by $y = \sqrt{x}$ above and the $x$-axis below.
- $x = 2$ to $x = 4$, where the region is bounded by $y = \sqrt{x}$ above and the line $y = x - 2$ below.

**Region 1: From $x = 0$ to $x = 2$**

In this region, the area is bounded above by $y = \sqrt{x}$ and below by the x-axis ($y = 0$).

The area for this part is

$$\begin{aligned} A_1 &= \int_0^2 \sqrt{x} dx \\ &= \left[\frac{2}{3}x^{\frac{3}{2}}\right]_0^2 \\ &= \frac{2}{3}\left[2^{\frac{3}{2}} - 0^{\frac{3}{2}}\right] \\ &= \frac{2}{3} \times 2\sqrt{2} \\ &= \frac{4\sqrt{2}}{3} \end{aligned}$$

**Region 2: From $x = 2$ to $x = 4$**

In this region, the area is bounded above by $y = \sqrt{x}$ and below by $y = x - 2$.

The area for this part is

$$\begin{aligned} A_2 &= \int_2^4 \left(\sqrt{x} - (x-2)\right) dx \\ &= \int_2^4 \left(\sqrt{x} - x + 2\right) dx \\ &= \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{1}{2}x^2 + 2x\right]_2^4 \\ &= \left(\frac{2}{3} \cdot 4^{\frac{3}{2}} - \frac{1}{2} \cdot 4^2 + 2 \cdot 4\right) - \left(\frac{2}{3} \cdot 2^{\frac{3}{2}} - \frac{1}{2} \cdot 2^2 + 2 \cdot 2\right) \\ &= \left(\frac{2}{3} \cdot 8 - 8 + 8\right) - \left(\frac{2}{3} \cdot 2\sqrt{2} - 2 + 4\right) \\ &= \left(\frac{16}{3}\right) - \left(\frac{4\sqrt{2}}{3} + 2\right) \\ &= \frac{10 - 4\sqrt{2}}{3} \end{aligned}$$

**Step 4: Add the Areas**

Finally, sum the areas $A_1$ and $A_2$.

$$\begin{aligned} A &= A_1 + A_2 \\ &= \frac{4\sqrt{2}}{3} + \left(\frac{10}{3} - \frac{4\sqrt{2}}{3}\right) = \frac{10}{3} \end{aligned}$$

Thus, the area of the region is $\frac{10}{3}$ sq. unit.

2. Show that if you cut a spherical loaf of bread into slices of equal width, each slice will have the same amount of (crust) outer surface area. [3]

**Solution:** Let the radius of the sphere be $r$. The sphere is generated by revolving the curve

$$y = \sqrt{r^2 - x^2}.$$

Let the width of the bread be $h$. It is cut down from the point $a$ to $a + h$. The surface area of the cut-down part is

$$\begin{aligned}
Crust &= \int_a^{a+h} 2\pi\sqrt{r^2 - x^2}\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx \\
&= \int_a^{a+h} 2\pi\sqrt{r^2 - x^2}\sqrt{1 + \frac{1}{2}\left(\frac{-2x}{\sqrt{r^2 - x^2}}\right)^2}\,dx \\
&= \int_a^{a+h} 2\pi\sqrt{r^2 - x^2}\left(\frac{r}{\sqrt{r^2 - x^2}}\right)dx \\
&= \int_a^{a+h} 2\pi r\,dx \\
&= 2\pi r h
\end{aligned}$$

3. Consider the improper integral 8

$$\int_1^\infty \frac{\cos t}{t} dt.$$

Show that the improper integral is convergent. Discuss about the absolute convergence of this improper integral.

**Solution: Dirichlet test** Let $f, g : [a, \infty) \to \mathbb{R}$ be such that

1. $f(x)$ is decreasing and $f(x) \to 0$ as $x \to \infty$,
2. $g(x)$ is continuous and there exists $M > 0$ such that $\int_a^t g(x)dx \leq M$ for all $t > a$.

Then $\int_a^\infty f(x)g(x)dx$ converges.

Here $f(t) = \frac{1}{t}$ and $g(t) = \cos t$ with $M = 2$.

By the Dirichlet test, the improper integral converges. But it is not absolutely convergent. For, (One can do the interval shift so that cos has a single wave)

$$\begin{aligned}
\int_1^{2\pi k} \left|\frac{\cos x}{x}\right| \, dx &\geq \sum_{n=1}^{k-1} \int_{2\pi n}^{2\pi(n+1)} \left|\frac{\cos x}{x}\right| \, dx \\
&\geq \sum_{n=1}^{k-1} \frac{1}{2\pi(n+1)} \int_{2\pi n}^{2\pi(n+1)} |\cos x| \, dx \\
&= \sum_{n=1}^{k-1} \frac{1}{2\pi(n+1)} \int_0^{2\pi} |\cos x| \, dx \\
&= \sum_{n=1}^{k-1} \frac{2}{\pi(n+1)}.
\end{aligned}$$

As $\sum_{n=1}^\infty \frac{2}{\pi(n+1)}$ is diverges, hence the improper integral diverges. That is, the improper integral $\int_1^{+\infty} \frac{\cos x}{x} dx$ converges, but not absolutely!

4. Consider the function 3

$$f(x,y) = \begin{cases} \frac{5x^7y^8}{x^{21}+y^{12}} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Calculate $\lim_{(x,y)\to(0,0)} f(x,y)$. Justify your answer.

**Solution:** Consider the path $y^4 = mx^7$

$$f(x,y) = \frac{5x^7m^2x^{14}}{x^21 + m^3x^{21}} = \frac{5m^2}{1+m^3}$$

The limit depends on the path. Hence by two path test the limit do not exists.

5. Consider the function 8

$$f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Show that the function is continuous at every point in $\mathbb{R}^2$, using the $\epsilon - \delta$ definition.

**Solution:** At the point $(x,y) = (0,0)$

$$\left| \frac{3x^2y}{x^2+y^2} - 0 \right| = \frac{3x^2|y|}{x^2+y^2} \leq 3|y|$$

Choose, $\delta = \frac{\epsilon}{3}$ be given then $0 < \sqrt{x^2+y^2} < \delta$ implies

$$\left| \frac{3x^2y}{x^2+y^2} - 0 \right| \leq 3\sqrt{x^2+y^2} < 3\delta = \epsilon$$

Therefore $f$ is continuous at $(0,0)$.

For other points, using the algebraic properties of continuity with numerator and denominator are continuous, the function is continuous.

6. Show that at every point $(x_0, y_0)$ in the domain of a differentiable function $f(x, y)$, the gradient of $f$ is normal to the level curve through $(x_0, y_0)$. [3]

**Solution:** If a differentiable function $f(x, y)$ has a constant value $c$ along a smooth curve $r(t) = g(t)\mathbf{i} + h(t)\mathbf{j}$, then $f(g(t), h(t)) = c$. Differentiating both sides, we get

$$\frac{d}{dt} f(g(t), h(t)) = \frac{d}{dt}(c)$$
$$\frac{\partial f}{\partial x}\frac{dg}{dt} + \frac{\partial f}{\partial y}\frac{dh}{dt} = 0$$
$$\left(\frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j}\right) \cdot \left(\frac{dg}{dt}\mathbf{i} + \frac{dh}{dt}\mathbf{j}\right) = 0$$

That is, $\nabla f$ is normal to the tangent vector $\frac{dr}{dt}$, so it is normal to the curve. That is, at every point $(x_0, y_0)$ in the domain of a differentiable function $f(x, y)$, the gradient of $f$ is normal to the level curve through $(x_0, y_0)$

7. A flat circular plate has the shape of the region $x^2 + y^2 \leq 1$. The plate, including the boundary where $x^2 + y^2 = 1$, is heated so that the temperature at the point $(x, y)$ is [6]

$$T(x, y) = 4x^3 - 15x^2 + 12x - 3y^2.$$

Find the temperatures at the hottest and coldest points on the plate.

**Solution:** (Maxima and Minima) Critical points:

$$\frac{\partial T}{\partial x} = 12x^2 - 30x + 12 = 0 \Rightarrow x = \frac{1}{2} \text{ or } x = 2$$
$$\frac{\partial T}{\partial x} = -6y = 0 \Rightarrow y = 0$$

Hence the critical points are $(\frac{1}{2}, 0)$ and $(2, 0)$

$$\frac{\partial^2 T}{\partial x^2} = 24x - 30, \quad \frac{\partial^2 T}{\partial x \partial y} = 0, \quad \frac{\partial^2 T}{\partial y^2} = -6$$

The critical point $(\frac{1}{2}, 0)$ is inside the domain. At that point,

$$\frac{\partial^2 T}{\partial x^2} = 24\left(\frac{1}{2}\right) - 30 = -18 < 0,$$

$$\frac{\partial^2 T}{\partial x^2}\frac{\partial^2 T}{\partial y^2} - \frac{\partial^2 T}{\partial x \partial y} = (-18).(-6) - 0 = 108 > 0$$

Hence the point $(\frac{1}{2}, 0)$ is a local maximum and maximum value is

$$4\left(\frac{1}{2}\right)^3 - 15\left(\frac{1}{2}\right)^2 + 12\left(\frac{1}{2}\right) - 3(0)^2 = \frac{4}{8} - \frac{15}{4} + \frac{12}{2} = \frac{4 - 30 + 48}{8} = \mathbf{\frac{22}{8}}$$

The critical point $(2, 0)$ **is not in the domain** (saddle!).

On the boundary by Lagrange method, $\lambda = -3$ and we get points $(1, 0)$ and $(-1, 0)$ and at these points $T(1, 0) = 1$ and $T(-1, 0) = -31$.

The circular plate has maximum temperature of $\frac{11}{4}$ at $\left(\frac{1}{2}, 0\right)$ and minimum temperature of $-31$ on the boundary at $(-1, 0)$.

8. Using Lagrange's multiplier method, find the rectangular box of the largest volume that has sides parallel to the $x, y$ and $z$ axes and that can be inscribed in the ellipsoid given by $$5x^2 + 3y^2 + 7z^2 = 1.$$ [6]

**Solution:** Notice the rectangle inscribed in the ellipsoid intersect at $x$ then the corresponding side is $2x$ similarly for $y$ and $z$. Hence the problem becomes

$$\text{Maximize } f(x, yz) = 2x2y2z = 8xyz$$
$$\text{such that } g(x, y, z) = 5x^2 + 3y^2 + 7z^2 = 1$$

$\nabla f = \lambda \nabla g \Rightarrow (8yz, 8xz, 8xy) = \lambda(10x, 6y, 14z)$

Case (i) $\lambda = 0$ Then at least two of the coordinated are zero. Hence Volume is zero

Case(ii) $\lambda \neq 0$ then all are non-zero.

$$\frac{8yz}{8xz} = \frac{y}{x} = \frac{\lambda 10x}{\lambda 6y} = \frac{5x}{3y}$$

Similarly

$$\frac{8yz}{8xy} = \frac{5x}{7z}$$

That implies,

$$y^2 = \frac{5}{3}x^2, \quad z^2 = \frac{5}{7}x^2$$

Now $g = 0$ gives $5x^2 + 3y^2 + 7z^2 - 1 = 15x^2 - 1 = 0$

Which implies

$$x = \pm\frac{1}{\sqrt{3}\sqrt{5}}; \qquad y = \pm\frac{1}{\sqrt{3}\sqrt{3}}; \qquad z = \pm\frac{1}{\sqrt{3}\sqrt{7}}.$$

Hence the maximum is

$$f\left(\frac{1}{\sqrt{3}\sqrt{5}}, \frac{1}{\sqrt{3}\sqrt{3}}, \frac{1}{\sqrt{3}\sqrt{7}}\right) = \frac{8}{9\sqrt{35}}$$